科學 HW6

$u = t^z,\ du = z t^{z-1} dt$

$dv = e^{-t} dt,\ v = -e^{-t}$

11. #7. $\Gamma(z+1) = \int_0^\infty e^{-t} t^{z+1-1} dt = \int_0^\infty e^{-t} t^z dt.$

By Integration by part, $\Gamma(z+1) = -t^z e^{-t}\Big|_0^\infty + z\int_0^\infty e^{-t} t^{z-1} dt = 0 + z\int_0^\infty e^{-t} t^{z-1} dt = z\Gamma(z)$ #

$\left(\because \lim_{t\to\infty} -\frac{t^z}{e^t} \underset{L'H}{\overset{(\frac{\infty}{\infty})}{=}} \cdots = \lim_{t\to\infty} -\frac{z!}{e^t} = 0\right)$

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11. #9.

$U_{n+1} = U_n + h\left[(1-\frac{\alpha}{2}) f(x_n, U_n) + \frac{\alpha}{2} f(x_{n+1}, U_{n+1})\right] \to (*)$

① $\alpha = 0 \Rightarrow U_{n+1} = U_n + h f(x_n, U_n) \Rightarrow$ Forward Euler

② $\alpha = 1 \Rightarrow U_{n+1} = U_n + \frac{h}{2}(f_n + f_{n+1}) \Rightarrow$ Trapezoidal method.

③ $\alpha = 2 \Rightarrow U_{n+1} = U_n + h f(x_{n+1}, U_{n+1}) \Rightarrow$ Backward Euler.

Consider $y(x_{n+1}) = y(x_n) + h\left[(1-\frac{\alpha}{2}) y'(x_n) + \frac{\alpha}{2} y'(x_{n+1})\right]$

$= y(x_n) + h\left[(1-\frac{\alpha}{2}) y'(x_n) + \frac{\alpha}{2}(y'(x_n) + h y''(x_n))\right] + O(h^3)$

$\therefore\ y(x_{n+1}) = y(x_n) + h y'(x_n) + \alpha \frac{h^2}{2} y''(x_n) + O(h^3)$, then comparing the Taylor expansion, which is $y(x_{n+1}) = y(x_n) + h y'(x_n) + \frac{h^2}{2} y''(x_n) + O(h^3) \Rightarrow$ hence for $\alpha = 1$, it's 2nd-order consistent; and for $\alpha \neq 1$, we only have 1st-order consistent.

For $\alpha = 1$, $\begin{cases} y'(x) = -10y(x),\ x > 0 \\ y(0) = 1 \end{cases}$, $U_{n+1} = U_n + \frac{h}{2}(f_n + f_{n+1})$

$= U_n + \frac{h}{2}(f(x_n, y_n) + f(x_{n+1}, y_{n+1}))$.

$\therefore$ when $y'(x) = -10y(x) = f(x,y)$ $\Rightarrow U_{n+1} = U_n + \frac{h}{2}(-10U_n - 10U_{n+1})$

$\Rightarrow (1+5h)U_{n+1} = (1-5h)U_n \Rightarrow U_{n+1} = \frac{1-5h}{1+5h} U_n$

Hence, for the I.V.P, $\lambda = -10$, $\therefore Re(-10) < 0$, then consider the trapezoid method.

From the scheme $U_{n+1} = U_n + h\left[(1-\frac{\alpha}{2})\lambda U_n + \frac{\alpha}{2}\lambda U_{n+1}\right]$

$\Rightarrow U_{n+1}\left[1 - \frac{\alpha}{2} h\lambda\right] = U_n\left[1 + (1-\frac{\alpha}{2})h\lambda\right] \Rightarrow \frac{U_{n+1}}{U_n} = \frac{(1+(1-\frac{\alpha}{2})h\lambda)}{1-\frac{\alpha}{2}h\lambda}$

Take $\alpha = 1$, $\lambda = -10$, $\frac{U_{n+1}}{U_n} = \frac{1-5h}{1+5h} \Rightarrow U_n = \left(\frac{1-5h}{1+5h}\right)^n$, We want $\left|\frac{1-5h}{1+5h}\right| < 1$.

- For $h > 0$, $|1-5h| < 1+5h$ is holding. so for $h > 0$, this method for the I.V.P is absolutely stable #